

Shuyao Zhang

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EDUCATION EXPERIENCE

- 09/2026 – Now** National University of Singapore, Singapore. **Research Fellow**
- 09/2022 – 06/2026** Tianjin University, China. **Ph.D.**, Chemical Process Machinery. Supervisor: Prof. Xu Chen.
- 05/2025 – 05/2026** Nanyang Technological University, Singapore. **Visiting PhD**. Supervisor: Prof. Kun Zhou.
- 09/2016 – 06/2020** Tianjin University, China. **B.Eng.**, Process Equipment and Control Engineering. GPA: 3.53/4.0.
- 09/2019 – 12/2019** Carleton University, Canada. **Exchange program** under the Excellent Undergraduate Scholarship.
- 07/2019 – 08/2019** Duke University, USA. **Summer Scholar** Program. Final grade: A.

PUBLICATIONS

Total: 14 peer-reviewed papers (5 first / co-first author + 2 in preparation).

- 1) **Zhang S**, Dai H, Li Y, et al. Additive manufactured corrosion-resistant SS316L/IN625 functionally graded multi-material in hydrofluoric acid (HF) environment. Corrosion Science, 2024, 230: 111926.
- 2) **Zhang S**, Li Y, Zhao Y, et al. Elucidating corrosion discrepancies between directed energy deposited 316L stainless steel and Inconel 625 via micro-droplet characterization and electrochemical analysis in hydrofluoric acid (HF) vapor environment. Electrochimica Acta, 2025: 147357.
- 3) **Zhang S**, Li Y, Dai H, et al. Achieving balanced uniform and local corrosion performance of SS 316L/IN625 hybrid alloy manufactured by directed energy deposition. Journal of Materials Research and Technology, 2024, 32: 3937-3948.
- 4) Dai H, **Zhang S** (Co-first author), Li Y, et al. Stress corrosion cracking behavior of 316L manufactured by different additive manufacturing techniques in hydrofluoric acid vapor. Journal of Materials Science & Technology, 2024, 191: 33-48.
- 5) Yao J, **Zhang S** (Co-first author), Chen S, et al. Competitive adsorption and unexpected corrosion inhibition effect of 316L stainless steel in NaClO₃ and NaCl environment of chlor-alkali industry. Corrosion Science, 2026: 113777.
- 6) **Zhang S**, Cheng Zhang. Multi-ion governed passivation and breakdown of 316L stainless steel in NaCl/NaClO₃/NaOH alkaline environments. Corrosion Science. (Under Review)
- 7) **Zhang S**, Zhao, Niu, P., & Chen, X. TiC-enabled second-phase engineering in DED 316L stainless steel for simultaneous strength and corrosion resistance enhancement. (In Preparation)
- 8) **Zhang S**, Li, B, & Zhou, K. Heat-treatment-induced corrosion mode transition in DED-fabricated nickel-aluminium bronze: from uniform corrosion to selective dissolution. (In Preparation)
- 9) Ran Y, **Zhang S**, Zhao Z, et al. Ratcheting fatigue behavior and microstructural evolution of T91 steel under oxygen-controlled lead-bismuth eutectic environments[J]. Engineering Fracture Mechanics, 2026: 112404.
- 10) Li Y, **Zhang S**, Yu W, et al. Multiaxial low cycle fatigue behavior and life prediction of wire arc additive manufactured 308L stainless steel. International Journal of Fatigue, 2024, 183: 108241.
- 11) Zhao Y, **Zhang S**, Zhu J, et al. Built-in electric field-driven interfacial charge transfer for boosted pseudocapacitance. Chemical Engineering Journal, 2025: 166061.

- 12) Zhao Y, Gao Z, **Zhang S**, et al. Asymmetric-Charge-Distributed Co-Mn Diatomic Catalyst Enables Efficient Oxygen Reduction Reaction. *Advanced Functional Materials*, 2025, 35(37): 2504260.
- 13) Li Y, Dai H, **Zhang S**, et al. The enhanced high-temperature oxidation resistance of additively manufactured GH4169 by adding small amounts of 304L. *Journal of Materials Research and Technology*, 2024, 30: 164-173.
- 14) Guo, C., Shi, S., Yu, J., **Zhang, S**, et al. Addressing the strength-corrosion tradeoff in 316L stainless steel by introducing cellular ferrite via directed energy deposition. *Additive Manufacturing*, 2024, 86: 104201.

RESEARCH INTERESTS

Materials degradation in complex environments; interfacial electrochemistry and passivation; electrochemical diagnostics and multi-scale characterization; experimental platform and device design for degradation studies.

SKILLS AND EXPERTISE

- **Corrosion and electrochemistry:** electrochemical workstations (CHI, Gamry, Bio-Logic); potentiodynamic and potentiostatic polarisation, EIS, electrochemical noise; micro-droplet cell; in-situ stress-corrosion coupled testing system.
- **Metal additive manufacturing:** directed energy deposition (DED); laser powder bed fusion (LPBF).
- **Microstructure characterisation:** SEM (student supervisor for FEI SEM, Tianjin University Instrument Center), EDS, EBSD, TEM, XRD, FIB, HR-DIC.
- **Simulation and data analysis:** ABAQUS (elastoplastic and fatigue simulations); Python (NumPy, Pandas, scikit-learn); MATLAB; COMSOL Multiphysics.
- **Languages:** Mandarin (native); English (TOEFL 100).

CONFERENCE PRESENTATIONS

- Oral, International Symposium on Structural Integrity (ISSI), 2022.
- Oral, International Conference on Corrosion Protection and Application, Wuhan, 2023.
- Oral, PhD Forum on Chemical Engineering, Tianjin University, 2024.